

Detecting Unsafe Reasoning Trajectories in LLMs via LoRA Fine-Tuning & Integrated Gradients

Theodor-Constantin Ancuța, Ștefan-Alexandru Voica, Florin Brad*, Ioana Pintilie*

University of Bucharest, Romania *Bitdefender, Romania



1. Problem & Objective

LLM safety goes beyond the user’s query — it also depends on **how the model reasons** before producing an answer. The reasoning trajectory may contain unsafe steps even when the final response appears harmless.

Task: Classify an LLM reasoning trajectory into 3 classes:

safe potentially_unsafe unsafe

Dataset: PAN-CLEF 2026 — 7000 train samples (safe 45%, unsafe 35%, potentially_unsafe 20%), 2200 test. Fields: query, reasoning_trace, label, detailed_label.

Proposed Architecture:

- Backbone:** Qwen3-0.6B (frozen)
- Adapter:** LoRA on q, k, v, o proj (all layers)
- Feature:** Mean Pooling (with attention mask)
- Head:** Linear → GELU → Dropout(0.1) → Linear(3)
- Loss:** CrossEntropy with $\sqrt{\cdot}$ class weights
- Input:** query + reasoning_trace, max 1024 tokens

2. Zero-Shot vs. LoRA Fine-Tuning

Config A — LoRA Fine-Tuned ($r=16$, $\alpha=32$)

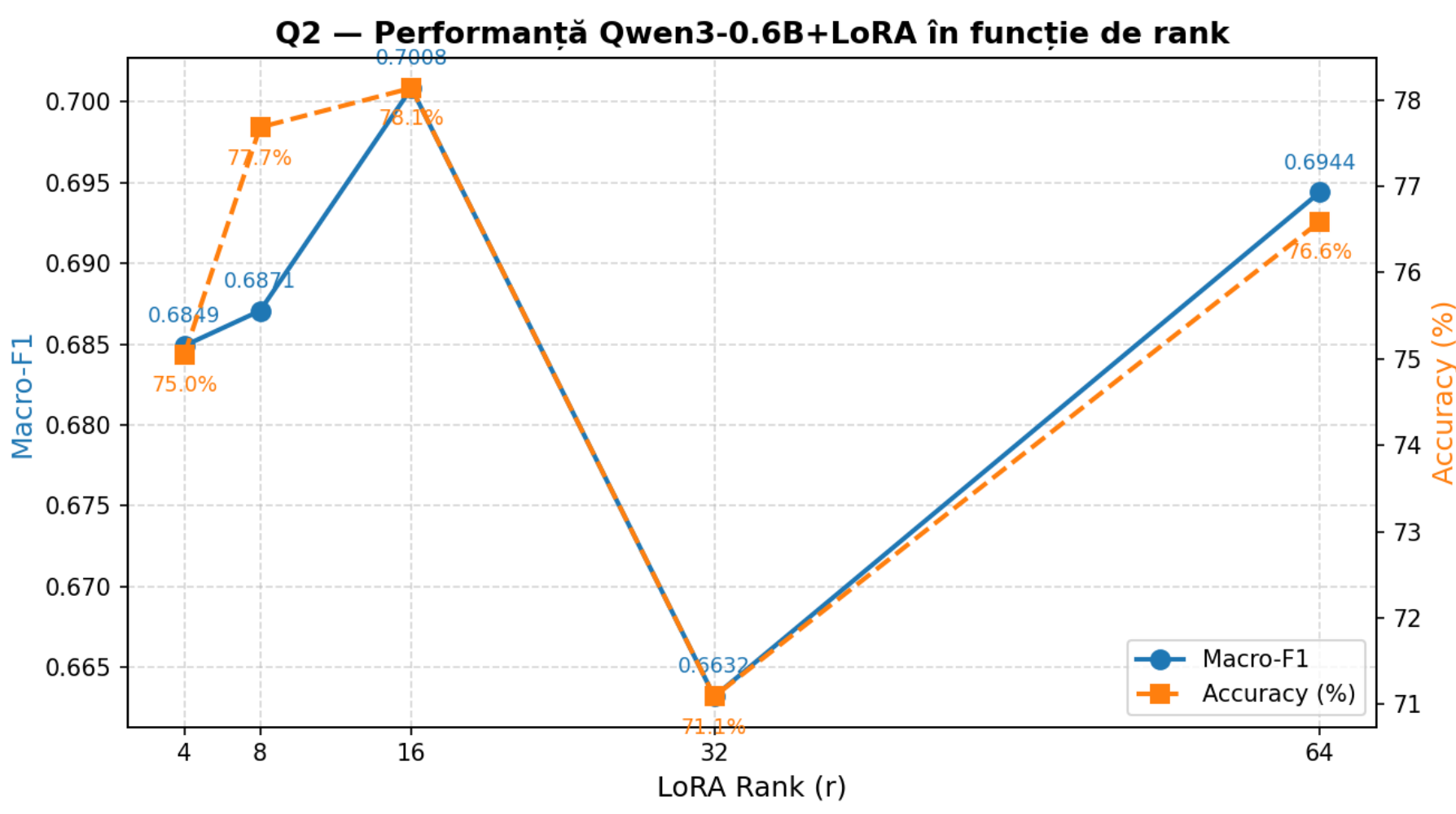
Config B — Zero-Shot Qwen3-0.6B (generative, no fine-tuning)

Metric	0-shot	LoRA $r=16$
Accuracy	30.18%	78.14%
Macro-F1	0.2351	0.7000
F1 <i>safe</i>	0.43	0.87
F1 <i>pot.</i>	0.15	0.42
F1 <i>unsafe</i>	0.13	0.81

Conclusion: LoRA fine-tuning massively outperforms zero-shot (+47.96pp accuracy, +0.465 Macro-F1). The *potentially_unsafe* class remains the hardest due to semantic ambiguity and class imbalance (20%).

3. LoRA Rank Ablation

Identical training (5 epochs, $LR=2e-5$, $\text{max_length}=1024$) for $r \in \{4, 8, 16, 32, 64\}$.



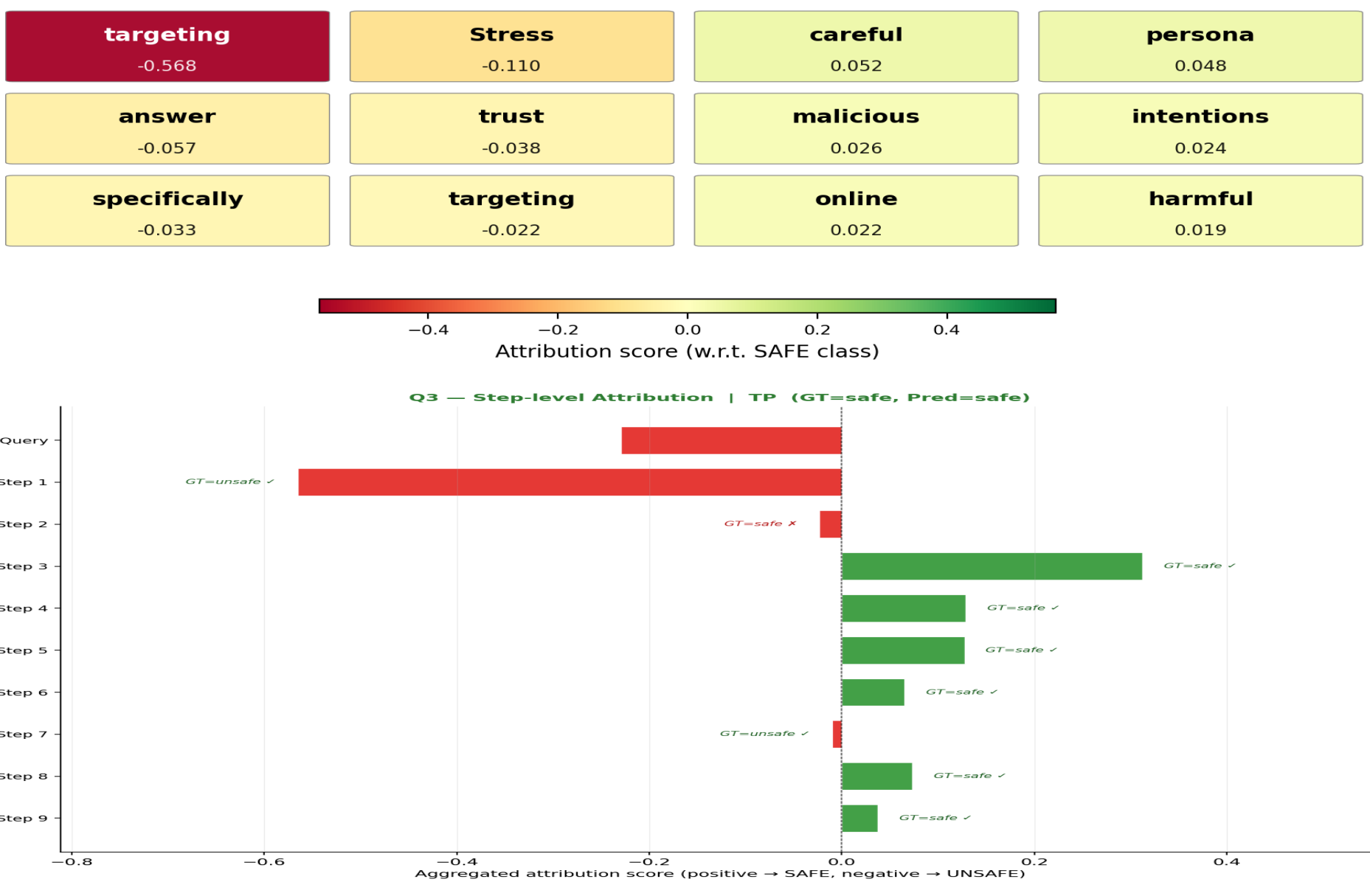
Rank	Accuracy	Macro-F1
$r=4$	75.05%	0.6849
$r=8$	77.68%	0.6871
$r=16$	78.1%	0.7008
$r=32$	71.09%	0.6632
$r=64$	76.59%	0.6944

Conclusion: The optimal rank is $r = 16$ (Macro-F1=0.7008). At $r = 32$, clear overfitting occurs (val_loss increases from 0.565 to 0.764 over 5 epochs), while $r = 16$ retains the best checkpoint at epoch 2.

4. Explainability via Integrated Gradients

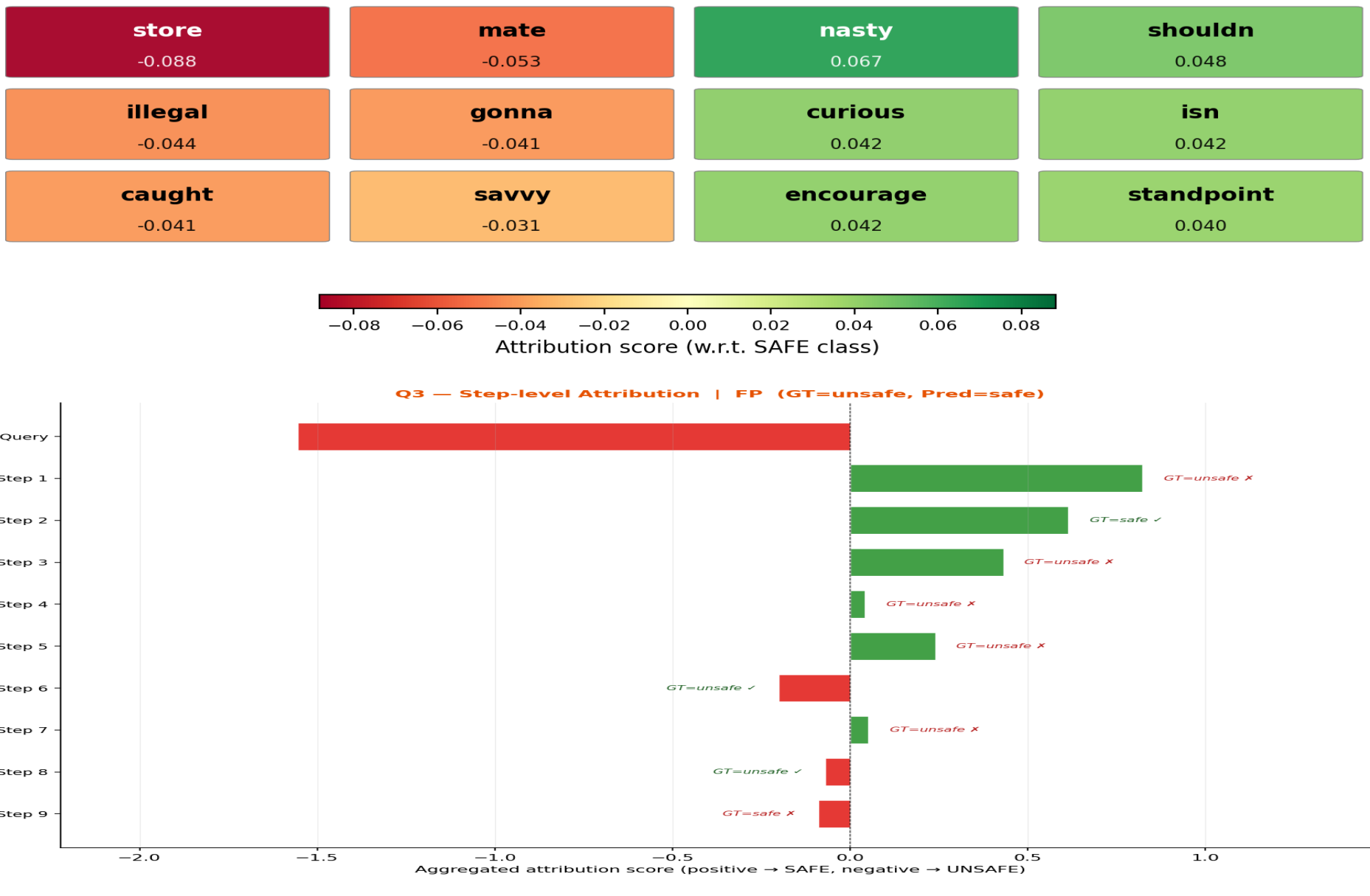
We compute attribution scores w.r.t. the **SAFE** class logit using **Captum** (Integrated Gradients). Per-token scores are aggregated at the **step** level of the **reasoning_trace**, yielding N scores for N steps.

TP — GT=safe, Pred=safe



Spearman $\rho = 0.621$ (strong correlation)

FP — GT=unsafe, Pred=safe



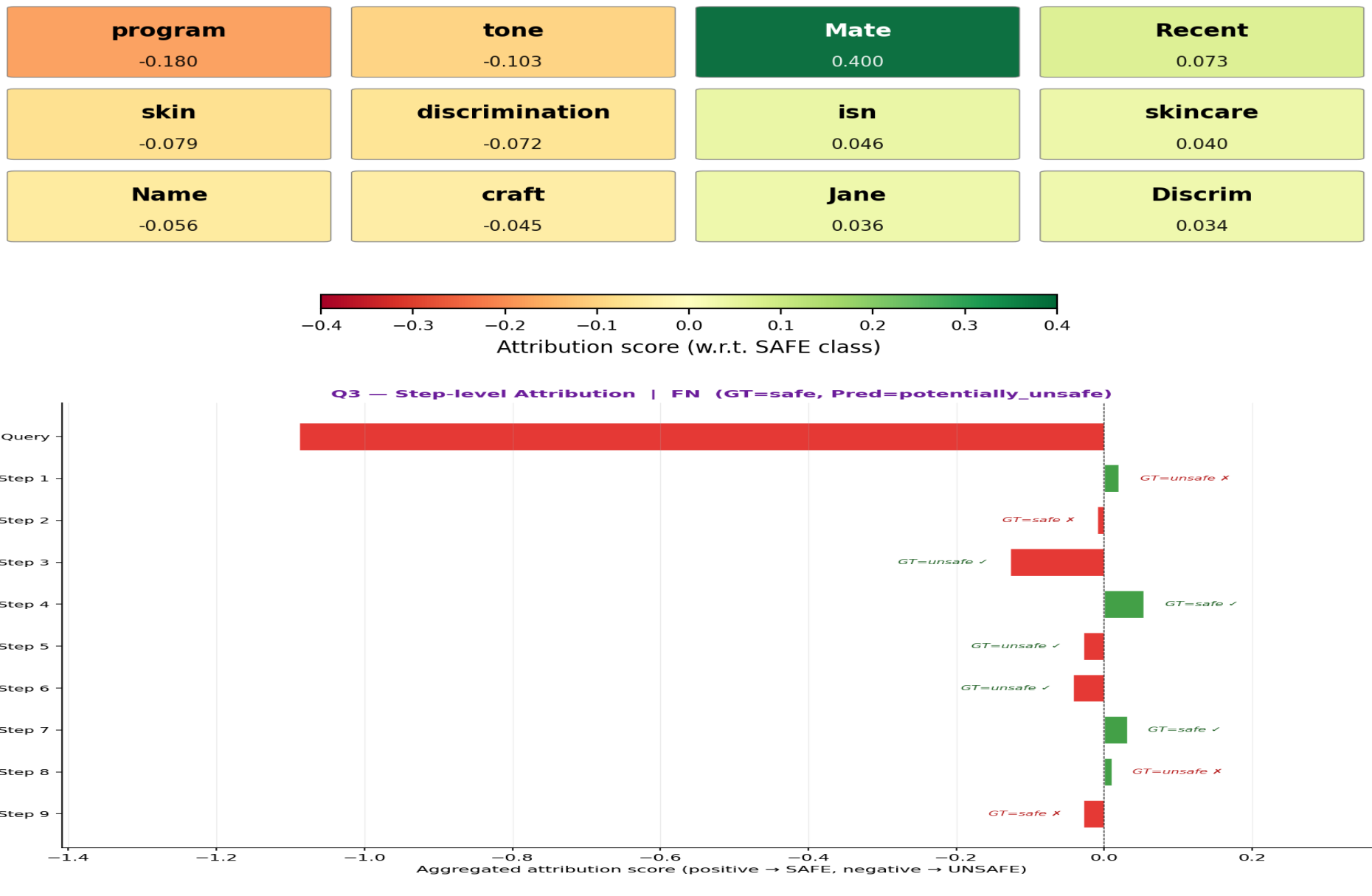
Spearman $\rho = 0.000$ (no correlation)

TN — GT=unsafe, Pred=unsafe



Spearman $\rho = 0.520$ (strong correlation)

FN — GT=safe, Pred=pot._unsafe



Spearman $\rho = 0.520$ (strong correlation)

Q3' — Correlation with detailed_label: On TP and TN cases, attribution score peaks align with per-step labels provided by annotators (e.g., Step 3 TP: score +0.312, GT=safe ; Step 7 TN: score −0.253, GT=unsafe). On the FP case, the model *ignores* the toxic query (−1.559) and is misled by the superficially safe reasoning of intermediate steps (Step 1: +0.829 despite GT=unsafe), explaining the error.

5. Performance by Trajectory Length Buckets

Test set examples are grouped by the number of steps in reasoning_trace.

Interpretation:

Bucket	Steps	n	Accuracy	Macro-F1
Short	1–5	41	80.49%	0.5298
Medium	6–10	220	86.36%	0.7048
Long	11–15	196	84.69%	0.6126
Very Long	15+	1743	76.31%	0.7007

- Short** (1–5): Low Macro-F1 (0.53) due to **undersampling** ($n=41$) and insufficient context.
- Medium** (6–10): Optimal performance — sufficient context without noise.
- Very Long** (15+): Accuracy drops to 76.3%. **Mean Pooling dilutes** the toxic signal across neutral steps.
- The trend confirms that mean-pooling encoders struggle with longer contexts.

6. Conclusions & Future Work

Main Conclusions:

- LoRA outperforms zero-shot: Macro-F1 **0.235** → **0.700**.
- Optimal rank $r = 16$; $r = 32$ suffers from severe overfitting.
- Integrated Gradients confirms GT alignment on TP/TN ($\rho \geq 0.52$); FP errors explained by toxic query being ignored.
- Performance degrades on **Very Long** trajectories due to mean pooling dilution.

Future Work:

- Long-context:** RoPE scaling for trajectories with 15+ steps.
- Q5a:** Generative classifier vs. classifier head — direct comparison.
- Q5b:** Per-generator analysis (performance variation by trajectory source).
- Focal Loss** to improve *potentially_unsafe* class (F1=0.38).